

Experiment 14 – Data-Driven Testing Using Robot Framework

1. Module Name and Experiment Title

Module Name: Module 4 – Robot Framework

Experiment Number: 14

Experiment Title: Data-Driven Testing Using Robot Framework

Programming Language: Python (Robot Framework)

Website: The Internet – Login Page

Website URL: <https://the-internet.herokuapp.com/login>

2. Problem Statement

To implement data-driven testing using Robot Framework by testing a login page with multiple sets of credentials. The automation should verify successful login with valid credentials and display appropriate error messages for invalid usernames and passwords.

3. Objective

- To understand data-driven testing in Robot Framework.
- To execute multiple test cases using different sets of login credentials.
- To use Robot Framework's test templates and reusable keywords.
- To validate successful and unsuccessful login attempts.
- To generate execution reports and logs.

4. Tools, Software and Concepts Used

Tools and Software:

- Python
- Robot Framework
- SeleniumLibrary
- Google Chrome
- Visual Studio Code
- The Internet login demo website

Concepts Used:

- Data-driven testing
- Test templates
- Robot Framework keywords
- Selenium browser automation
- Assertions and validation
- HTML reports and execution logs

5. Source Code / Implementation Steps

The experiment uses Robot Framework's data-driven testing approach to execute the same login procedure with different credentials.

test_data_driven.robot code:

*** Settings ***

Library SeleniumLibrary

Suite Setup Open Test Browser

Suite Teardown Close Browser

Test Template Verify Login With Different Credentials

*** Variables ***

\${URL} https://the-internet.herokuapp.com/login

\${BROWSER} chrome

\${SCREENSHOT_DIR} C:/Users/rajar/Desktop/Selenium/screenshots

*** Test Cases ***

Valid Login

tomsmith SuperSecretPassword! You logged into a secure area! 01_valid_login.png

Invalid Username

wronguser SuperSecretPassword! Your username is invalid!

02_invalid_username.png

Invalid Password

tomsmith wrongpassword Your password is invalid! 03_invalid_password.png

*** Keywords ***

Open Test Browser

Open Browser about:blank \${BROWSER}

Maximize Browser Window

Verify Login With Different Credentials

[Arguments] \${username} \${password} \${expected_message} \${screenshot}

Go To \${URL}

Wait Until Element Is Visible id=username timeout=10s

Input Text id=username \${username}

Input Text id=password \${password}

Click Button css=button[type="submit"]

Wait Until Page Contains \${expected_message} timeout=10s

Page Should Contain \${expected_message}

```
Capture Page Screenshot    ${SCREENSHOT_DIR}/${screenshot}  
Log    Login test completed for username: ${username}
```

Implementation Steps:

1. Import SeleniumLibrary in the Robot Framework test file.
2. Define the login page URL and browser configuration.
3. Create a reusable keyword to open the browser and navigate to the login page.
4. Create a login keyword that accepts a username and password.
5. Define a test template to reuse the login procedure.
6. Provide three test cases:
 - o Valid Login
 - o Invalid Username
 - o Invalid Password
7. Verify the expected result for each test case.
8. Execute the test file and generate the HTML report and log.

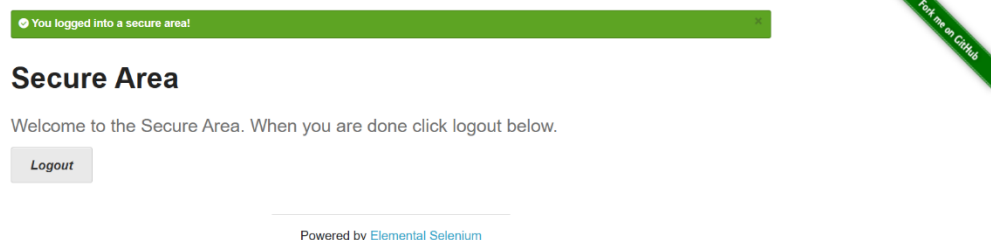
Note: Insert the exact contents of your `test_data_driven.robot` file in this section to keep the report consistent with your implementation.

6. Output with Clearly Labelled Screenshots

The test cases were executed successfully using Robot Framework. The generated report shows **3 tests passed, 0 failed, and 0 skipped**.

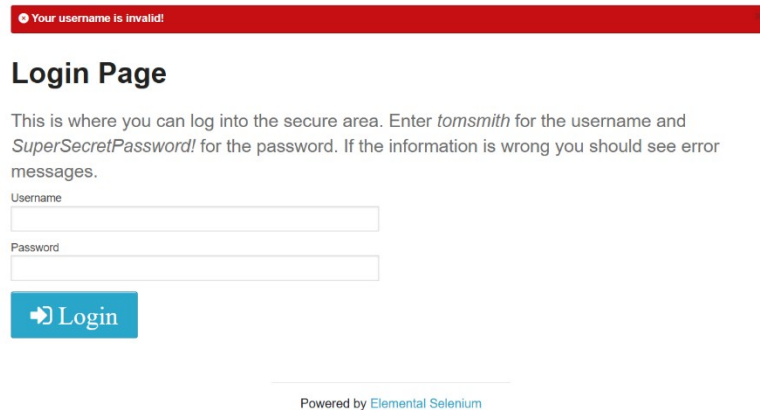
Screenshot 1: Successful Login

Shows the secure area after entering valid login credentials. The success message confirms that the login was successful.



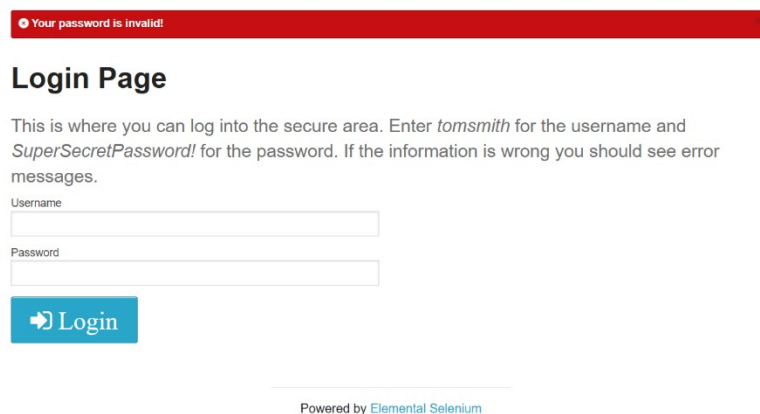
Screenshot 2: Invalid Username

Shows the login page displaying the error message “Your username is invalid!” after an invalid username was entered.



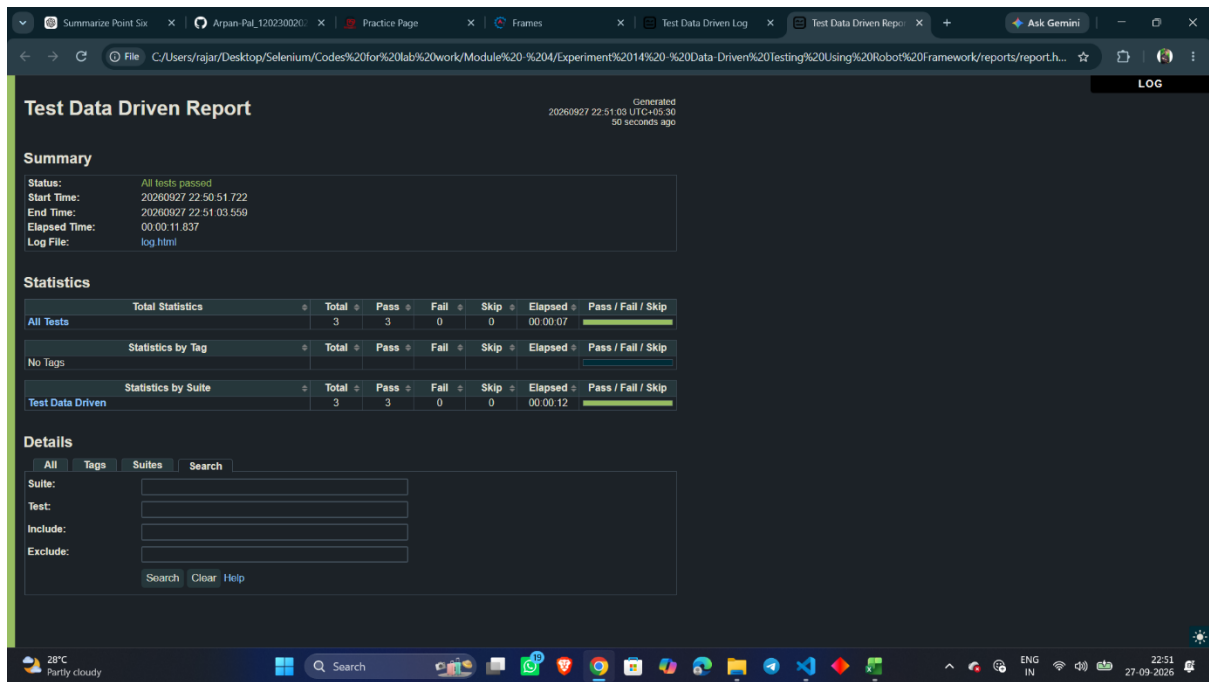
Screenshot 3: Invalid Password

Shows the login page displaying the error message “Your password is invalid!” after an invalid password was entered.



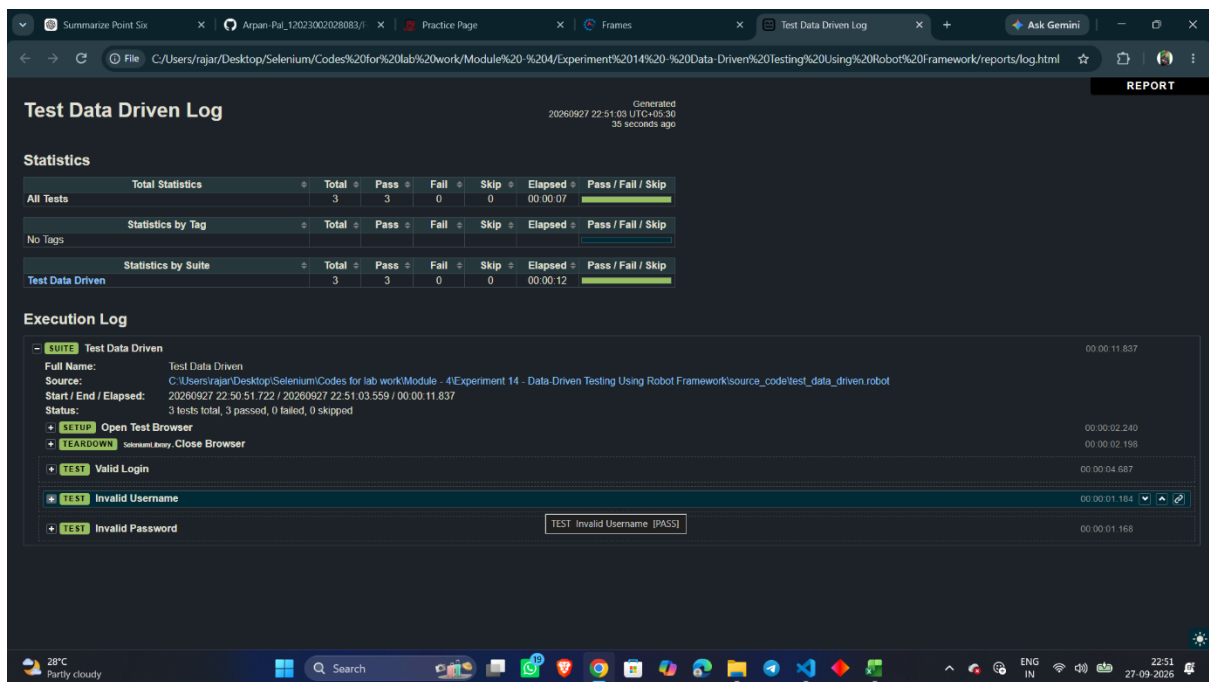
Screenshot 4: Robot Framework Test Report

Shows the execution summary with 3 total tests, 3 passed, 0 failed, and 0 skipped.



Screenshot 5: Robot Framework Execution Log

Shows the execution details for the three test cases: Valid Login, Invalid Username, and Invalid Password.



7. Result, Observation and Conclusion

The experiment was successfully completed using Robot Framework and SeleniumLibrary. Data-driven testing was implemented to test the login functionality with different sets of credentials.

The execution report confirmed that all three test cases passed. The application allowed login with valid credentials and displayed the expected error messages for invalid credentials.

This experiment demonstrated how Robot Framework's data-driven testing features can reduce code duplication and help validate application behavior with multiple input combinations.